

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Agentic AI for Mental Health Awareness & Suicide Prevention

Domain of Project – Healthcare

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Problem Statement -

Problem Statement: Agentic AI for Mental Health Awareness & Suicide Prevention

The Challenge

Mental health issues such as depression, anxiety, and suicidal thoughts are growing worldwide. Stigma, lack of awareness, and limited access to timely help prevent many individuals from seeking support. Traditional systems are often reactive, leaving gaps in early detection and preventive care.

The Objective

Build an Agentic AI-powered Mental Health Awareness & Suicide Prevention Agent that is empathetic, proactive, and safe. The solution should:

1. Promote Awareness & Education – Share accessible resources on mindfulness, coping strategies, and self-care.
2. Enable Early Detection (RAG) – Use Retrieval-Augmented Generation to identify distress signals in conversations, journaling inputs, or safe online data.
3. Provide Empathetic Support – Build a conversational agent that listens actively and responds with empathy (with clear disclaimers: not a substitute for professional medical advice).
4. Predict & Prevent Risks – Forecast stress peaks, suggest preventive strategies, and issue alerts when high-risk signals are detected (with consent).
5. Connect to Human Support – Link users to helplines, counselors, or community support for timely intervention.

Tech Stack (Mandatory)

- LangFlow + IBM watsonx.ai + IBM Granite Models

Technology Used

I used following technology

1. LangFlow
2. IBM watsonx.ai
3. IBM Granite Models
4. Meta LLAMA model

Proposed Solution: Agentic AI for Mental Health Awareness & Suicide Prevention

The proposed solution is a **multi-agent AI-powered Mental Health Awareness & Suicide Prevention System** built using **LangFlow**, **IBM watsonx.ai**, and **IBM Granite Models** to provide empathetic, proactive, and personalized mental health support. The system addresses challenges such as lack of awareness, delayed intervention, and limited access to timely help through an intelligent and modular architecture.

At its core, the solution utilizes a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve reliable information from trusted mental health resources, mindfulness guides, coping strategy documents, and crisis support materials. The retrieved knowledge is embedded and stored in a vector database, enabling Granite models to generate accurate and context-aware responses.

The system consists of three key agents:

- **MindGuide AI (Awareness & Education Agent)** provides information on stress, anxiety, depression, mindfulness, self-care practices, and healthy coping strategies while promoting mental health awareness and reducing stigma.
 - **Sentinel AI (Risk Detection Agent)** analyzes user conversations and journaling inputs to identify emotional distress, detect early warning signs of anxiety, depression, or suicidal ideation, classify risk levels, and recommend preventive actions using RAG-based reasoning.
 - **HopeBridge AI (Empathetic Support Agent)** offers compassionate conversations, personalized coping strategies, wellness plans, and motivational guidance while encouraging users to seek professional help or connect with helplines when high-risk situations are detected.
- The solution provides an interactive chat interface where users can ask questions, record journal entries, and receive personalized support. With user consent, the system can monitor emotional patterns, forecast stress peaks, and generate preventive recommendations to improve overall well-being.

LangFlow orchestrates the multi-agent workflow, **IBM watsonx.ai** provides scalable AI infrastructure and governance, while **IBM Granite Models** power retrieval, reasoning, summarization, and empathetic response generation.

Overall, the proposed solution delivers an intelligent mental health assistant that promotes awareness, enables early detection of emotional distress, provides empathetic support, predicts potential risks, and connects users to appropriate human assistance, fostering proactive and preventive mental healthcare.

Langflow component Used -

1. **Chat Input** – Allows users to enter mental health queries or journal entries for AI processing.
2. **IBM watsonx.ai** – Processes user input using IBM Granite models to generate intelligent and context-aware responses.
3. **IBM Granite-4.0-h-small Model** – Provides fast and efficient reasoning for mental health awareness and conversational support.
4. **MindGuide AI Agent** – Delivers mental health education, self-care tips, and awareness-based guidance.
5. **Sentinel AI Agent** – Detects emotional distress and assesses potential mental health risks from user conversations.
6. **HopeBridge AI Agent** – Offers empathetic support, coping strategies, and encourages professional help when needed.
7. **Read File Component** – Reads the uploaded mental health PDF and provides knowledge for AI responses.
8. **Prompt Template Component** – Dynamically combines user queries with document content to create context-aware prompts.
9. **Language Model Component** – Generates responses based on the prepared prompt and retrieved knowledge.
10. **Chat Output** – Displays the final personalized mental health response to the user.

Langflow workflow -

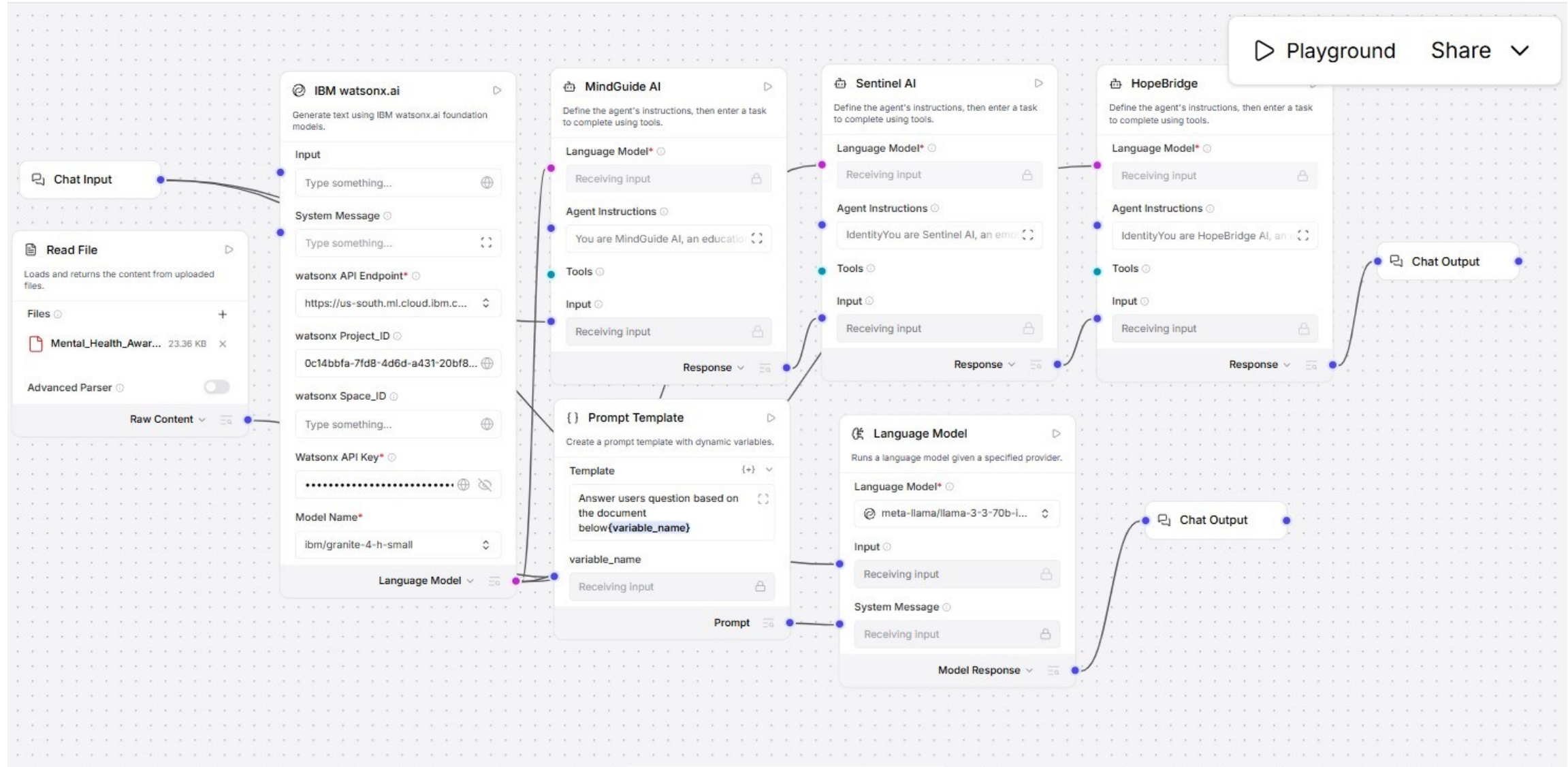
Starter Project /  Mental Agent

 150k

 25k



 Playground  Share 



API generation

The screenshot shows the IBM Cloud IAM API keys management interface. A modal window titled "API key successfully created" is displayed in the center. The modal contains the following text:

API key successfully created

Copy the API key or click download to save it. You won't be able to see this API key again, so you can't retrieve it later. The API key is no longer displayed after 273 seconds.

API key

.....

Copy [icon] Download [icon]

The background interface shows the "API keys" section in the left sidebar. The main content area displays a table of API keys. The first key is highlighted:

API key	Expires in
cpd-apikey-IBMid-692001OLP1-2026-06-11T10:21:02Z	6-11-2026 10:21 GMT

The bottom of the screen shows a Windows taskbar with the date and time 16:27 11-06-2026.

Project_id generation

The screenshot displays the IBM Watsonx web interface for managing a project named 'mental agent'. The browser address bar shows the URL: `dataplatform.cloud.ibm.com/projects/0c14bbfa-7fd8-4d6d-a431-20bf8c6cd7fc/manage/general?context=wx`. The interface includes a top navigation bar with the IBM Watsonx logo and user account information (Fatema Vhora's Account, Dallas). Below this, a breadcrumb trail shows 'Projects / mental agent'. The main content area is divided into tabs: Overview, Assets, Deployments, Jobs, and Manage (which is currently selected). On the left, a sidebar lists various project settings: General (selected), Access control, Environments, Services & integrations, Tools, and Pipeline. The 'General' tab displays the following information:

- Name:** mental agent
- Description:** What's the purpose of this project?
- Tags:** Add tags to make projects easier to find.
- Project ID:** 0c14bbfa-7fd8-4d6d-a431-20bf8c6cd7fc

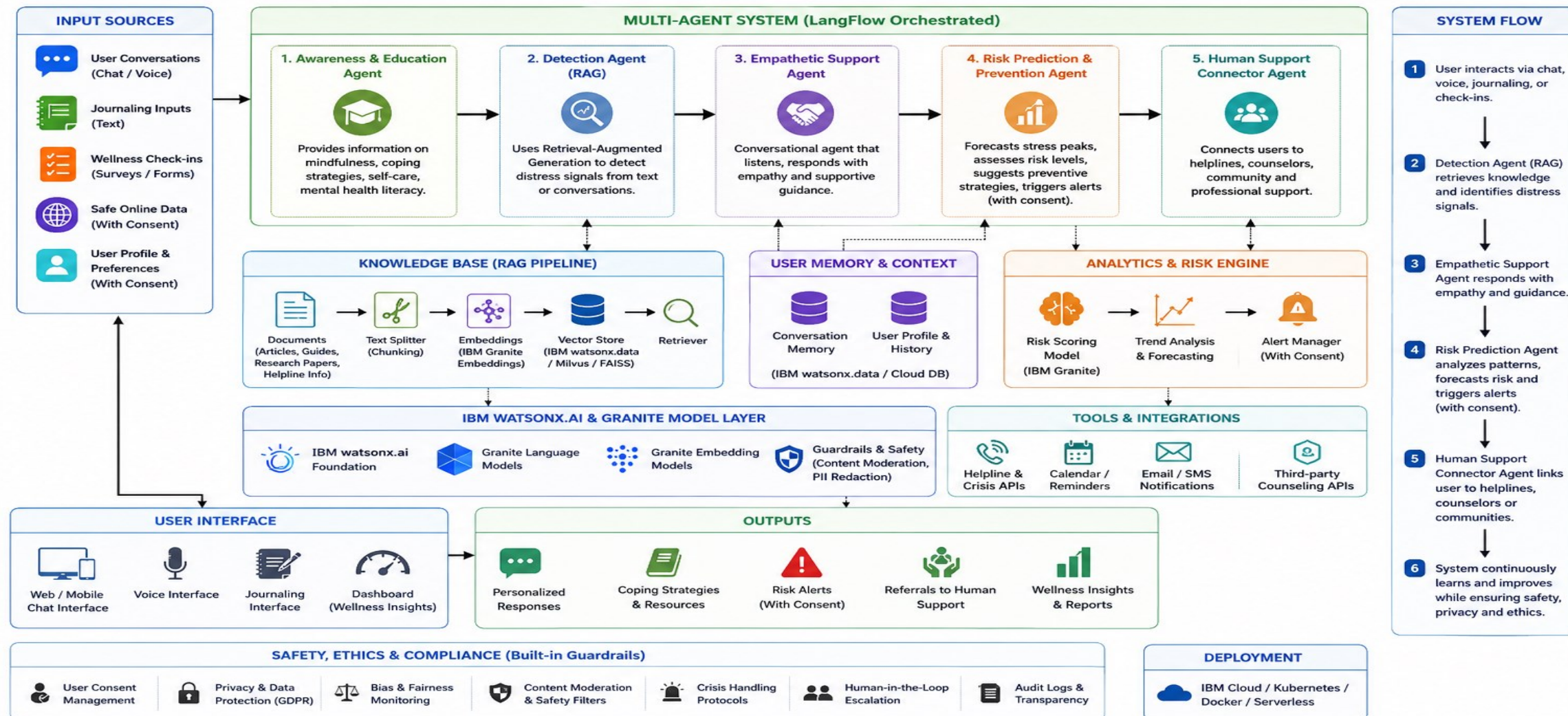
On the right side of the 'General' tab, there is a 'Storage used' section showing 0 Bytes, a 'Bucket' section with the name 'mentalagent-donotdelete-pr-560cr3vcmbkbo', and a 'Regenerate credentials' section with a 'Regenerate' button. Below these is a 'Manage in IBM Cloud' link. At the bottom, a 'Notifications' section is partially visible.

The Windows taskbar at the bottom shows the system clock as 16:30 on 11-06-2026, with a temperature of 38°C and weather 'Mostly sunny'.

Architecture blueprint

Agentic AI for Mental Health Awareness & Suicide Prevention Model Blueprint Architecture

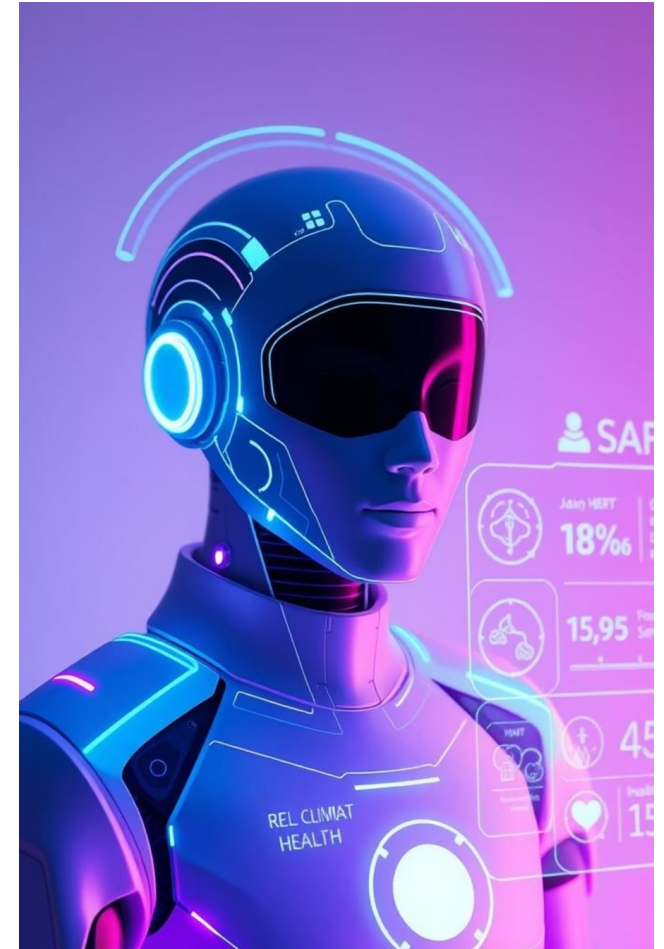
Objective: Build an empathetic, proactive, and safe AI agent that promotes awareness, enables early detection, provides support, predicts risks, and connects users to human help.



Role of Agentic AI in the solution

Role of Agentic AI in Mental Health Awareness & Suicide Prevention Agent

- Agentic AI enables the Mental Health Awareness & Suicide Prevention Agent to function as an intelligent, proactive support system rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized agents to provide personalized mental health guidance and early intervention.
- Each agent performs a dedicated role—**MindGuide AI** promotes mental health awareness and education, **Sentinel AI** analyzes conversations to detect emotional distress and suicide risk, and **HopeBridge AI** provides empathetic support, coping strategies, and encourages professional help when needed. This collaborative multi-agent approach ensures accurate and efficient assistance.
- Agentic AI also provides context awareness by understanding users' emotions, concerns, and conversation history to generate personalized recommendations. It can proactively identify warning signs, suggest preventive measures, and adapt its responses based on the user's emotional state.
- Additionally, the agents communicate and share insights with each other, enabling better decision-making and delivering comprehensive support while maintaining safety and empathy.
- Overall, Agentic AI makes the system intelligent, context-aware, and user-centric, acting as a compassionate digital mental health companion that promotes awareness, supports early detection, and facilitates timely intervention.



Technology Used

- **LangFlow Platform** – Used for designing, orchestrating, and managing the multi-agent workflow visually, enabling seamless interaction between the mental health awareness, risk detection, and support agents.
- **IBM Granite Model (Granite-4.0-h-small)** – Powers natural language understanding, reasoning, emotion-aware conversations, and personalized mental health guidance with fast and efficient response generation.
- **IBM Cloud** – Provides a secure, scalable, and reliable infrastructure to build, deploy, and manage the Mental Health Awareness & Suicide Prevention Agent.
- **Retrieval-Augmented Generation (RAG)** – Retrieves trusted mental health knowledge and coping strategies from uploaded documents to generate accurate, context-aware, and evidence-based responses.
- **IBM watsonx.ai** – Supports foundation model deployment, prompt execution, AI governance, and integration with Granite models for intelligent conversational capabilities.
- **Knowledge Base (PDF/File Reader)** – Stores and retrieves mental health awareness documents, self-care guides, and educational resources that enrich AI responses with domain-specific knowledge.

Token Usage

1,27,611 token are used

SessioDay 3MorniService

m xResouLangfiYou aLangfiAPI kAPI kIBM CGoogl+

dataplatform.cloud.ibm.com/projects/0c14bbfa-7fd8-4d6d-a431-20bf8c6cd7fc/overview?context=wx

New Chrome available

IBM watsonx

Upgrade?

Fatema Vhora's Account

Dallas

FV

Projects / mental agent

View all

Collapse

Jump back in

By all

Assets that you create with tools show here. See all assets, including data assets, on the Assets page.



[View all](#)

Resource usage

For this month in this project

0

CUH

127611

Tokens

0

Hosting hours

0

Pages

Your documentation

Get started with your documentation

You can create and manage documents about work that you do in this project.

[Open Documentation editor](#)

Project history

You created project mental agent

Today at 4:19 PM

4 38°C Mostly sunny

Search



ENG IN

17:48 11-06-2026

Chat Output Screenshot

Note: Here for every prompt 2 output will be generated because one response will be from PDF document and another will be from pre feeded agent information

Agent Response

The screenshot shows a web browser window with the address bar displaying `localhost:7860/flow/ce2d68b4-ad21-44cb-8aeb-4435af57e66f`. The browser's tab bar includes several open tabs, and the address bar shows a notification for 'New Chrome available'. The main content area displays a chat interface for 'Mental Agent' under the 'Starter Project'.

On the left side of the chat interface, there is a sidebar with a 'Sessions' list. The list includes 'Default Session', 'New Session 0', 'New Session 1', and 'New Session 2'. 'New Session 2' is currently selected and highlighted.

The chat history shows a conversation between a 'User' and an 'AI' agent. The user's message is: 'Can you connect me to the human support for my anxiety issue?'. The AI's response is: 'I'm so sorry to hear that you're struggling with anxiety. I'm here to provide you with support and guidance, but I'm a large language model, I'm not a substitute for human support. If you're in need of immediate support, I recommend reaching out to a crisis helpline or a mental health professional. Here are some resources that may be helpful:'. Below this text is a bulleted list of resources: 'National Alliance on Mental Illness (NAMI) Helpline: 1-800-950-6264', 'Crisis Text Line: Text "HOME" to 741741', and 'Substance Abuse and Mental Health Services Administration (SAMHSA) Helpline: 1-800-662-HELP (4357)'. The AI's response concludes with: 'You can also search online for mental health resources in your area, such as therapists, support groups, and counseling services. Remember, you're not alone, and there are people who care about you and want to help. Please take'.

At the bottom of the chat interface, there is a text input field with a microphone icon and a send button (an upward arrow).

The Windows taskbar at the bottom of the screen shows the system clock as 17:36 on 11-06-2026, the language set to ENG IN, and the weather as 38°C Mostly sunny.

Sessions

Default Session

☐ Select All

New Session 0

New Session 1

New Session 2

New Session 2

- **Call 911** (or your local emergency number) right away.
- **National Suicide & Crisis Lifeline:** Dial **988** in the United States (or use the chat at 988lifeline.org). They offer 24/7 confidential support.
- **Local crisis lines:** Search for "crisis hotline" plus your city or region for a phone number that's staffed by trained counselors.

Ongoing support options:

- **Therapy & counseling:** Look for licensed therapists in your area through directories such as Psychology Today, GoodTherapy, or your insurance provider's network. Many therapists now offer tele-therapy if in-person visits are difficult.
- **Support groups:** Organizations like the Anxiety and Depression Association of America (ADAA) and local community centers often run group sessions—both in person and online.
- **Self-help resources:** Evidence-based apps (e.g., Headspace, Calm, Woebot) can teach coping skills such as mindfulness, deep breathing, and cognitive-behavioral techniques. Books on managing anxiety (e.g., *The Anxiety and Phobia Workbook* by Edmund Bourne) are also helpful.

If you'd like more specific recommendations—such as finding a therapist near you, locating a local support group, or learning about particular coping strategies—just let me know what details would be most useful (e.g., your location, preferred format of help). I'm here to point you toward the right resources so you can start feeling better soon.

4 38°C
Mostly sunny

Search

ENG
IN

17:37
11-06-2026

Agent Response

SessioDay 3MorniIdentimentaResouLangflYou aiL xAPI k API kIBM CGoogl+New Chrome available

localhost:7860/flow/ce2d68b4-ad21-44cb-8aeb-4435af57e66f

Starter Project / Mental Agent150k25k

Sessions+
Default Session
☐ Select All
New Session 0
New Session 1
New Session 2

New Session 2

User

Can you connect me to the human support for my anxiety issue?

AI

Finished in 18.5K | 4.0s

I'm so sorry to hear that you're struggling with anxiety. I'm here to provide you with support and guidance, but I'm a large language model, I'm not a substitute for human support. If you're in need of immediate support, I recommend reaching out to a crisis helpline or a mental health professional.

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Remember, you're not alone, and there are people who care about you and want to help. Please take

4 38°C Mostly sunny

Search

Google

Excel

PowerPoint

ENG IN

17:36 11-06-2026

SessioDay 3MorniIdentimentaResouLangfiYou aLi xAPI kAPI kIBM CGoogl+New Chrome available

localhost:7860/flow/ce2d68b4-ad21-44cb-8aeb-4435af57e66f

Starter Project / Mental Agent150k25k

Sessions+
Default Session
☐ Select All
New Session 0
New Session 1
New Session 2

New Session 2

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4 38°C
Mostly sunny

Search

Google

Word

PowerPoint

Paint

ENG IN

17:37
11-06-2026

Novelty and Uniqueness

The proposed **Mental Health Awareness & Suicide Prevention Agent** stands out by combining **Agentic AI**, **Retrieval-Augmented Generation (RAG)**, and **empathetic intelligence** into a single proactive support system.

Multi-Agent Intelligence – Instead of relying on a single chatbot, three specialized agents (MindGuide AI, Sentinel AI, and HopeBridge AI) collaborate to provide education, risk detection, and emotional support.

RAG-Based Knowledge Retrieval – Integrates trusted mental health resources with AI-generated responses, ensuring accurate, evidence-based, and context-aware guidance while reducing misinformation.

Early Risk Detection – Analyzes user conversations and journal entries to identify emotional distress, anxiety, depression, and potential suicide risk, enabling timely preventive intervention.

Empathetic & Personalized Support – Adapts responses based on the user's emotional state and conversation history, delivering compassionate guidance and tailored coping strategies.

Proactive Prevention – Goes beyond answering questions by forecasting stress patterns, suggesting wellness routines, and recommending preventive actions before crises escalate.

Human-in-the-Loop Assistance – Connects users with helplines, counselors, or trusted support systems when high-risk situations are detected, ensuring responsible AI assistance.

Scalable & Enterprise-Ready – Built using LangFlow, IBM watsonx.ai, and IBM Granite Models, providing a secure, reliable, and scalable platform for real-world deployment.

This makes the solution an intelligent, adaptive, and proactive digital mental health companion rather than a traditional question-answering chatbot.

Git Hub Link

Github URL

https://github.com/fatema1804/FDP_FATEMA_AGENTIC_AI_EDUNET

Uploaded following three files into your github repository

- 1) Agentic AI for Mental Health Awareness & Suicide Prevention.json file ,
- 2) Problem statement.pdf file
- 3)FDP_Fatema_Agentic_AI_Edunet_Mental_health_awareness_prevention.pptx file

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Future Scope

1. Integration with Wearable Devices & Digital Health Platforms –

The system can be integrated with smartwatches and health monitoring devices to analyze physiological indicators such as heart rate, sleep patterns, and activity levels. This will enable the Mental Health Agent to detect stress trends, monitor emotional well-being, and provide more personalized preventive recommendations.

2. AI-Powered Voice Assistant & Multilingual Support –

Enhancing the solution with voice-based interaction and multilingual capabilities will improve accessibility for users from diverse backgrounds. Individuals can communicate naturally in their preferred language, making mental health support more inclusive, user-friendly, and accessible to a wider population.

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Thank You!

Thank you for your time and interest.